CSE 291 Project Plan

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Introduction and Problem Statement

Introduction

The goal of this project is to develop machine learning models to classify COVID-19 patient conditions using serum proteomics data from the Proteomic and Metabolomic Characterization of COVID-19 Patient Sera dataset. The dataset contains peptide abundance measurements across patient samples, and represents patients in the following four categories: healthy, symptomatic without COVID-19, have severe COVID-19, or have non-severe COVID-19. Rather than directly performing a single multiclass classification, we decompose the problem into multiple binary classification tasks: distinguishing healthy patients from infected ones (including both COVID cases and those symptomatic without COVID), symptomatic non-COVID cases from COVID cases, and severe COVID-19 cases from non-severe COVID-19 cases.

Problem Formulation

Each patient sample is represented as a d-dimensional feature vector, where each dimension corresponds to the measured abundance of a peptide variant. Using the Variants TMT peptide intensity table from this dataset, we have approximately n~90 patient samples and d~100K peptide variant features. The task is to learn a function that maps peptide abundance vectors (our input) to binary class labels (our output) with the following classification tasks:

- Healthy vs. infected
- Symptomatic non-COVID vs. COVID
- Severe vs. non-severe COVID

The objective is to learn models that accurately classify patient condition across the defined tasks, and generalize well to unseen samples. A secondary objective is for the model to identify a subset of peptide features that are consistently predictive of disease status and severity, as this will increase the interpretability of our results.

Input Definition

The input data comes from the peptide intensity matrix, where columns that have the suffix _intensity_for_peptide_variant correspond to abundance measurements for peptide variants. To construct out input vectors, only columns without the suffix _unmod are used, and we transpose the matrix so that each row corresponds to a patient and each column corresponds to a peptide variant feature. Each feature value represents the relative abundance (fold variation) of a peptide variant across samples. Therefore we get our input: high-dimensional numerical feature vectors of peptide variant abundances.

We note that abundance values are not directly comparable across different peptides, but are meaningful when comparing the same peptide across different samples. Metadata columns such as peptide sequence, charge, and protein mappings are excluded from the input feature vectors, but will be used for downstream interpretation of selected features.

This input definition works well with our proposed models since having each patient encoded as a high-dimensional numerical feature vector is a standard setup for tabular machine learning models. Additionally, linear models such as logistic regression and Linear SVM can perform well with high-dimensional data. Tree-based ensemble methods such as Random Forest and XGBoost can use this representation to capture nonlinear relationships and interactions between peptides.

Handling Missing Values

Since missing values are prevalent in the dataset, we adopt the following strategy to handle them:

- Convert zeros to NaN (missing values) before processing
- Remove features with excessive missingness (e.g. >50% missing values) as they are unlikely to provide reliable signals
- For retained features we will use the following and compare:
    - Simple imputation using mean or median
    - Imputing with small values (near lower intensity range)

Mean/median imputation provides a simple and stable baseline for handling missing values without introducing excessive variance in this small-sample setting. In addition, imputing missing entries with small values reflects the proteomics-specific assumption that many missing measurements correspond to peptides with abundances below the detection limit, thereby preserving biological plausibility. While linear models (e.g., logistic regression, SVM) require explicit imputation, tree-based models such as XGBoost can natively handle missing values, allowing us to compare imputed and non-imputed pipelines. Additionally, although removing features with high missingness improves robustness, it may discard biologically relevant low-abundance peptides; therefore, we will evaluate different missingness thresholds to balance signal retention and noise reduction.

Considering Feature Subsets

Another issue with the data is that its dimensionality of ~100K features is extremely high, especially when the low sample size of ~90 is taken into account. While it is true that linear methods can handle high dimensional data well, this is a significant imbalance, which increases the risk of overfitting and model instability. To help mitigate this, we can reduce the input space to a subset of informative features through feature selection. Our feature selection process is outlined in “Feature Selection Strategy” in section 2.

Proposed Approach

Model Selection

Given the characteristics of the COVID-19 serum proteomics dataset - specifically the small number of samples (~90) and very high dimensional feature space (~100K peptide variants), we propose to use a combination of regularized linear models and tree-based ensemble methods. These models are well-suited for tabular biological data, allow control of overfitting, and provide interpretable feature importance for downstream biological validation. We will evaluate the following four models:

Regularized Logistic Regression

- Regularized logistic regression serves as a strong baseline model for this task. It is particularly suitable for datasets with high dimensionality and limited sample size, where overfitting is a major concern.
- By applying L1 (Lasso) or Elastic Net regularization, the model can perform implicit feature selection, shrinking many coefficients to zero and retaining only the most informative peptide features.
- This is especially important in our setting, where only a small subset of peptides is expected to be truly associated with COVID status or disease severity.
- Furthermore, logistic regression provides directly interpretable coefficients, enabling us to rank peptides and connect them to proteins for downstream biological and protein-level validation.
- Optimization and Regularization:
    - Hyperparameters to tune:
        - Regularization type: L1, L2, Elastic Net
        - Regularization strength ©
        - Elastic Net mixing parameter (l1_ratio)
    - Optimization strategy:
        - Perform cross-validation (e.g., 5-fold) to select optimal hyperparameters
        - Use solvers such as liblinear or saga depending on regularization type
    - Overfitting control:
        - Strong regularization (small C)
        - Feature pre-selection (filtering top-k peptides based on statistical tests)

XGBoost (Gradient Boosted Decision Trees)

- XGBoost is a powerful ensemble learning method capable of capturing nonlinear relationships and interactions between peptide features. This is particularly relevant because individual peptides often show overlapping distributions across classes, while combinations of peptides may better distinguish between conditions.

- Given the hierarchical classification tasks (healthy vs infected, symptomatic non-COVID vs COVID, severe vs non-severe), different subsets and interactions of features may be important for each task. XGBoost is well-suited to automatically learn such patterns. Additionally, it provides feature importance scores, which can be used to identify candidate biomarkers.
- Optimization and Regularization
    - Hyperparameters to tune:
        - Number of trees (n_estimators)
        - Learning rate (eta)
        - Maximum tree depth (max_depth)
        - Subsampling ratios (subsample, colsample_bytree)
        - Regularization parameters (lambda for L2, alpha for L1)
    - Optimization strategy:
        - Grid search or randomized search with cross-validation
        - Early stopping using validation set
    - Overfitting control:
        - Use shallow trees (low max_depth)
        - Low learning rate with more trees
        - Subsampling of rows and features
        - Early stopping based on validation loss

Random Forest

- Random Forest is an ensemble method that constructs multiple decision trees and aggregates their predictions. It is robust to noise and can model nonlinear relationships and feature interactions without requiring extensive tuning. It serves as a useful intermediate model between linear models and more complex boosting methods.
- Random Forest also provides feature importance measures, allowing us to compare important peptides across different models and identify consistently relevant biomarkers.
- Optimization and Regularization
    - Hyperparameters to tune:
        - Number of trees (n_estimators)
        - Maximum depth of trees (max_depth)
        - Minimum samples per split/leaf (min_samples_split, min_samples_leaf)
        - Number of features considered per split (max_features)
    - Optimization strategy:
        - Cross-validation to select optimal parameters
    - Overfitting control:
        - Limit tree depth
        - Increase minimum samples per leaf
        - Use feature subsampling (max_features)

Linear Support Vector Machine (SVM)

- Linear SVM is well-suited for high-dimensional feature spaces, particularly when the number of features exceeds the number of samples. It aims to find a hyperplane that maximizes the margin between classes, which can be effective when classes are separable using a sparse subset of features.
- In our setting, after feature selection, linear SVM can serve as a strong alternative to logistic regression and help validate whether the learned decision boundary is robust.
- Optimization and Regularization
    - Hyperparameters to tune:
        - Regularization parameter C
        - Loss function (hinge or squared hinge)
    - Optimization strategy:
        - Cross-validation to select optimal C
    - Overfitting control:
        - Smaller C values for stronger regularization
        - Feature selection prior to training

Despite their strengths, the proposed models have limitations. Linear models such as logistic regression and SVM may fail to capture nonlinear interactions between peptides. Tree-based models, while flexible, are prone to overfitting in small-sample settings and may produce unstable feature importance rankings. Additionally, all models are sensitive to the high dimensionality and missing values in the dataset, making careful preprocessing and validation essential.

We will evaluate and compare these models across all classification tasks:

- Healthy vs. infected
- Symptomatic non-COVID vs. COVID
- Severe vs. non-severe COVID

Feature importance and model outputs will be used not only for prediction but also to identify candidate peptide and protein biomarkers, which will be further validated using statistical and biological criteria in Stage 2.

Feature Selection Strategy

Feature selection is essential for this project due to the inherent characteristics of the COVID-19 serum proteomics dataset. The dataset contains approximately ~100,000 peptide variant features but only ~90 samples, resulting in an extreme high-dimensional, low-sample-size (p >> n) setting.

In such scenarios:

- Models are highly prone to overfitting
- Many features are likely irrelevant or noisy
- Computational cost increases significantly
- Interpretability (critical for biological validation) becomes difficult

Additionally, the dataset contains a substantial number of missing values, as peptide intensities are not observed uniformly across all samples. As noted in the dataset description, missing values are common in mass spectrometry data due to detection limits and acquisition variability. These properties make feature selection not only reasonable but necessary for building robust and interpretable models.

Based on initial inspection of the dataset:

- Total number of features (peptide variants): ~100,000
- Number of samples: ~90
- Many peptide features contain missing values across multiple samples
- Some peptides are observed in only a small subset of samples

Key observations guiding feature selection:

- A large fraction of features have at least one missing value
- Some features have very high missingness (>50–80%)
- Many features show low variance across samples

These observations suggest that:

- Not all peptides contribute meaningful information
- Missingness and noise must be explicitly handled
- Dimensionality reduction is required before modeling

We propose a multi-stage feature selection pipeline combining filtering and model-based methods:

1. **Missing Value Filtering**
    - Remove peptide features with high missingness (>50% missing values across samples)
    - Features missing in most samples are unlikely to provide reliable signals
    - Reduces noise introduced by imputation
    - Improves stability of statistical tests and models

2. **Variance-Based Filtering**
    - Remove features with low variance across samples

- Features with near-constant values do not contribute to classification
- Helps eliminate uninformative peptides early

3. **Statistical Feature Selection**
   - Perform univariate statistical tests between classes:
     - Student's t-test (difference in means)
     - Mann-Whitney U test (difference in medians)
   - Select top-k features based on p-values
   - Reduces feature space to peptides that show statistically significant differences between groups

4. **Model-Based Feature Selection**
   - Use feature importance scores from:
     - Logistic Regression (magnitude of coefficients)
     - XGBoost / Random Forest (gain or impurity-based importance)
   - Captures multivariate relationships between features
   - Identifies peptides that are important in combination with others

5. **Combined Strategy**
   - The final feature set will be obtained by combining:
     - Statistically significant features (from univariate tests)
     - Features with high importance in trained models

Evaluation of the Project Models

Train, Validation, and Test Sets

Our dataset contains approximately 90 samples (90 patients), which is too small for a standard train/test split. Therefore, we will implement a 5-fold nested cross validation. In each fold, we will use stratified sampling to ensure that the distribution of the classes remains consistent across the training, validation, and test sets. Below is a description of the process:

- **Outer Loop:** The dataset will be divided into five folds with approximately 18 samples each. In each iteration, four of the folds (approximately 72 samples) will be used for training and validation, and the fifth set (approximately 18 samples) will be set aside for testing. Given that the four patient categories are relatively well-balanced, each test fold will contain about 4-5 samples per class (roughly a 25% distribution).
- **Inner Loop:** For each of the four folds that are selected for training, we will separate a portion to create a validation set (3 folds for training and 1 for validation). This inner loop is used to tune hyperparameters (such as the regularization strength for logistic regression or the tree depth for XGBoost) and to select the best-performing model (choosing between Lasso, Ridge, or Elastic Net) before testing it on the outer fold.

Scoring Metrics

To measure the success of our results, we will assess two factors: accuracy and reliability. To assess classification accuracy without being skewed by potential class imbalances, we will use balanced accuracy. To evaluate reliability and trustworthiness of the model's predictions, we will use F1-scores, which treats the identification of the COVID groups with equal weight to the healthy group. Then, we can take the average across the categories to get the overall performance.

As mentioned above, we also want to identify peptides that might be correlated with COVID-19. We do this by providing all the peptides to our model and having it predict the most relevant peptides. For each patient, these chosen peptides are used to calculate the probability that the patient has COVID. Then, we can plot all the patients' probabilities on an AUPRC curve to see how well the model performs at classifying the different groups, specifically distinguishing COVID cases from the non-COVID cases. We specifically chose AUPRC because it focuses on the model's ability to identify the COVID patients without being inflated by a possible high score on the health patients.

Performance Baseline

Even though our dataset is pretty small, the groups are pretty well-balanced, and there is less likelihood of group bias. However, we will still have a reference baseline to make sure our subgroups are accurately represented. This baseline represents pure probability, where we calculate the chance of randomly selecting each group given its proportion in the data. For our results to be successful, our models must demonstrate a significantly higher performance over this baseline.

Assessing Limitations and Alternative Approaches

Assessing Limitations

Given the characteristics of the Proteomic and Metabolomic Characterization of COVID-19 Patient Sera dataset, our proposed modeling approach faces several limitations:

- **Extreme High-Dimensionality and Overfitting:** We are dealing with approximately 90 patient samples (n) and ~100,000 peptide variant features (p). This severe p>>n scenario means our models, particularly the more complex tree-based ensembles (XGBoost, Random Forest), are at an extremely high risk of overfitting. They may memorize the noise in the training folds rather than learning generalizable biological signals.
- **Pervasive Missing Values (Sparsity):** Mass spectrometry data contains a high rate of missing values due to detection limits. While we plan to filter out features with >50% missingness and impute the rest, mean/median imputation on such a small sample size can reduce feature variance and introduce bias, potentially distorting the true biological distribution.
- **Small Sample Size and Subtask Imbalance:** Decomposing the problem into specific binary classification subtasks further reduces the available sample size for each specific model. This makes our evaluation highly sensitive to outliers.

Alternative Approaches

To address and reduce the impact of these expected limitations, we have established the following alternative fallback approaches:

- **Mitigating Overfitting:** If the model demonstrates severe overfitting, we will fall back exclusively to strongly regularized linear models. Lasso will force a sparse solution that restricts the model from utilizing too many peptides.
- **Alternative to Mean/Median Imputation:** If simple imputation introduces too much bias or noise, our alternative approach is to implement K-Nearest Neighbors (KNN) imputation. KNN imputation estimates missing peptide abundances based on the profiles of the most similar patients, which may better preserve the underlying biological relationships in the data compared to a global mean/median.
- **Addressing Sample Size Limitations:** If the dataset proves too small to yield stable results across the granular binary classification tasks, our alternative approach is to simplify the objective. Instead of granular classifications, we will consolidate the labels into a single, broader binary task: COVID-19 Positive vs. COVID-19 Negative. This consolidation will maximize the sample size per class, providing the models with a larger, more stable dataset to learn the primary infection signature before attempting severity classification.